Branton DeMoss

Contact

branton.demoss@maths.ox.ac.uk
www.brantondemoss.com
+44 (0)7926 576225

Mathematical Institute
Woodstock Road, Oxford
OX2 6GG, UK

Education

DPhil Machine Learning 2025
University of Oxford

BA Mathematics and Physics 2018
University of Colorado Boulder

Visitor Mathematical and Theoretical Physics 2016-17
University of Oxford

Experience

Mathematical Institute, University of Oxford 2025-
Postdoctoral Research Associate
- Research on the mathematical and computational foundations of AI.
- Erlangen AI Hub affiliate.

Oxford Robotics Institute 2021-25
Graduate Student Researcher
- Research in complexity, generalization, reinforcement learning, world models.

The Collaboratory 2020-23
Co-founder; Chief Science Officer
- Deep learning on language and graphs for knowledge curation.
- Led product strategy, design, and ML R&D.

Comma.ai 2020
ML Research Intern
- Reinforcement learning for self-driving cars.

Front Range Geosciences 2017-20
Machine Learning Engineer
- Sole dev on ML system for seismic data, sold to multinational co.

Center for Theory of Quantum Matter 2017
Research Assistant
- Research on quantum many-body localization.

Mathematics Department, CU Boulder 2016
Research Assistant
- Knot theory and topological quantum field theory.

High Enery Particle Physics Group, Physics Department, CU Boulder 2014-15
Research Assistant
- High performance Monte Carlo simulations (C++) for DUNE experiment.

Publications

Geometry, Generalization, and Effective Dimensions 2026
Work in progress.

Building on my earlier complexity work, I am proposing a unifying theory of phase transitions in neural networks, based on ideas from information geometry.

The Complexity Dynamics of Grokking 2025
Physica D: Nonlinear Phenomena

LUMOS: Language-Conditioned Imitation Learning with World Models 2024
ICRA 2025

DITTO: Offline Imitation Learning with World Models 2023
arχiv:2302.03086

Combining physics and deep learning to automatically pick first breaks in the Permian Basin 2021
First International Meeting for Applied Geoscience & Energy

Ein Liebesbrief an KataGo 2020
Deutsche Go Zeitung, Ausgabe 4/2020

Love Letter to KataGo, or: Go AI Past, Present, and Future 2020
American Go E-Journal

DeepTrace: A breakthrough application of deep learning to automate first break picking 2019
SEG 2019 Lenovo Thought Leadership Series

Topology and Knot Theory 2016
Course notes for CU Boulder special topics course:
"Topology, Knot Theory, and their applications in Physics and Chemistry"

Secondary Particle Showers from Hadron Absorber Interactions 2016
Deep Underground Neutrino Experiment (DUNE) Collaboration Documents

Teaching

Theoretical Physics for Machine Learning 2026
Oxford, HT 26

Physics of Information and Complexity 2024
Received highest possible marks for teaching performance.
Oxford, HT 24

Philosophy of Emergence 2024
Received highest possible marks for teaching performance.
Oxford, HT 24

Topics in Reinforcement Learning 2023
Received highest possible marks for teaching performance.
Oxford, MT 23

Rocket League Behaviour Cloning from Unlabelled Data 2023
Supervised Master's Thesis, Oxford
Student obtained highest marks, and secured funded DPhil position in Oxford.

Talks

Erlangen AI Hub 2025
Talk on information phase transitions in neural networks, and related gauge-theoretic issues.

2nd Symposium on Algorithmic Information Theory and Machine Learning 2025
Talk on my discovery of complexity phase transitions in learning systems. Link.

ICRA 2025, Robot Foundation Models Session 2025
Talk on our work LUMOS, addressing reinforcement learning in world models.

Harvard/Tufts, Levin Group 2025
Invited talk on complexity dynamics to Michael Levin's computational biology group. Link.

Oxford, Department of Physics 2024
Invited talk on complexity dynamics to Ard Louis's research group.

Oxford, Department of Statistics 2024
Invited talk on complexity and generalization to the RainML group.

Awards

AWS Lighthouse Scholarship (fully funded PhD)	Oxford, 2021
Stribic-Martin Scholarship ($10k for top mathematics undergrad)	Boulder, 2017
UROP Fellowship ($3k summer research grant)	Boulder, 2017
Dawkins Fund Award (Exeter College)	Oxford, 2016
Gilman Scholarship ($5k US Dept of State)	Oxford, 2016
Esteemed Scholar Award ($7k/yr for top enrolling undergrads)	Boulder, 2014

References

Prof. Nick Hawes
Professor of AI and Robotics, Oxford
Director, Oxford Robotics Institute
nickh@robots.ox.ac.uk

Prof. Ingmar Posner
Professor of Applied AI, Oxford
Deputy Director, Oxford Robotics Institute
Associate Head of Engineering Dept. (Research)
ingmar@robots.ox.ac.uk

Prof. Jakob Foerster
Associate Professor, Information Engineering, Oxford
jakob@robots.ox.ac.uk

Prof. Jared Tanner
Professor of the Mathematics of Information, Oxford
jared.tanner@maths.ox.ac.uk